Online Search Agents and Unknown Environments

Lesson emphasis

How online agents differ from offline planners, why unknown environments force interleaving of action and computation, and how performance and safety must be evaluated in the real world rather than only in an abstract search tree.

For class use

Each example in this deck is phrased for discussion, so you can pause and ask students to classify the situation, predict the agent's behavior, and defend whether a cautious or aggressive search policy is more rational.

Learning outcome and lecture direction

- Explain why online search is needed when an agent cannot rely on a complete map, complete transition knowledge, or a fully precomputed route before acting.
- Differentiate offline search from online search not only by timing, but by what the agent is assumed to know, what it must discover through action, and how real path cost accumulates during exploration.
- Analyze how performance is judged in unknown environments, including exploration overhead, revisits, backtracking, delayed commitment, and the possibility that a route found in hindsight may still have been expensive to discover.
- Evaluate safety considerations, especially when exploratory moves are costly, risky, irreversible, or operationally constrained, as in robotics, navigation, and real-world autonomous systems.

Core question for students

When the world is unknown, should an intelligent agent keep planning longer, or must it act in order to learn what can be planned at all?

Reminder

Do not confuse online search with partial observability. This is mainly about unknown global structure, not necessarily uncertainty about the current state itself.

Where this topic sits in our discussions

- We move from classical search to more realistic settings: nondeterministic actions, partial observability, and then online search. This progression matters because it shows that once the simplifying assumptions of classical search are relaxed, the form of rational problem solving must also change.
- By the time the chapter reaches online search, the issue is no longer just "Which node should the algorithm expand next?" The issue becomes "How should an agent behave when expanding the space requires physically moving through it?"
- This bridges classical search and later topics in learning and agent behavior by showing that an intelligent agent may need to discover the problem while solving it.

Roadmap

Classical search

Search with nondeterministic actions

Search in partially observable environments

Online search agents and unknown environments

Offline search versus online search

Dimension	Offline search	Online search
Knowledge before acting	Assumes enough model knowledge to compute a route in advance. The agent can reason over states it has not physically visited.	Starts with incomplete task-relevant knowledge. The agent must move, observe, and update its map while the task is unfolding.
Relationship between planning and action	Planning precedes execution. Deliberation can be extensive because no real move is made until a plan is chosen.	Planning and acting are interleaved. Each action both changes the world and reveals information relevant to later choices.
What counts as search cost	Node expansion is computational cost, but wrong hypothetical branches do not directly add path cost in the world.	Exploration itself is costly. A wrong turn, dead end, or backtrack is a real expenditure of time, energy, and opportunity.
Typical advantage	Can achieve stronger efficiency or optimality when the model is accurate and complete.	Can operate in unfamiliar environments where waiting for a complete model is impossible or unrealistic.

What “unknown environment” means here

- In this context, unknown does not simply mean that the problem is difficult. It means the agent lacks a complete task-relevant model of the environment before it starts acting. It may not know what states exist, which paths connect them, what certain actions cost, or which routes are blocked until it physically explores.
- This is different from partial observability as treated in the previous lessons In many online-search formulations, the agent can still identify the state it reaches after an action; what it does not know in advance is the **global structure of the world.**
- The practical consequence is that search becomes **map construction plus decision making**. The agent is not merely selecting from known alternatives; it is discovering what the alternatives are.

Example framing

A robot entering an unfamiliar campus building may know what “move forward” means locally, yet still not know the hallway layout, blocked exits, stair access, or travel cost to the goal.

Pedagogical caution

Do not overstate hidden-state uncertainty here. Emphasis is on unknown structure that must be discovered through experience.

How an online search agent operates

1. Perceive current state

The agent gets local information about where it is now and what immediate actions are available.

2. Update internal map

New observations are integrated into memory so the world model becomes incrementally less incomplete.

3. Choose next action

The agent balances progress toward the goal with the value and risk of exploring unfamiliar space.

4. Execute and learn

The chosen action changes the world and simultaneously reveals new structure, costs, or obstacles.

5. Repeat until termination

The process continues until the goal is reached, the search is abandoned, or no reasonable safe action remains.

This loop is why online search is better described as situated decision making than as one-shot route computation.

Example 1: Service robot in an unfamiliar academic building

- Suppose a delivery robot is asked to bring equipment from the lobby to a laboratory on the third floor of a building it has never mapped. It knows its current location and basic motion actions, but it does not know where the elevators are, whether corridors are blocked, or whether certain doors require access credentials.
- An offline planner would struggle because it lacks the transition graph needed to compute a full route before motion begins. The online agent, by contrast, can move through the building, remember discovered intersections, update what routes are traversable, and adapt when a hallway that looked promising ends in a locked door.
- This example is useful because students can immediately see that intelligence here is not just path selection. It is also efficient exploration, memory use, and prudent recovery from failed attempts.

Discussion prompts

- What information is known locally, and what must be discovered through movement?
- Which exploratory move seems rational if the robot wants progress but also wants to avoid being trapped in a service corridor?
- How would battery limits alter the agent's willingness to backtrack?

Example 2: Drone route finding after a storm

- Imagine a drone tasked with reaching a remote clinic after a storm. Pre-storm maps exist, but fallen trees, debris, wind corridors, and temporary no-fly zones make those maps unreliable. The agent therefore cannot treat the problem as a purely known shortest-path task.
- Online search becomes necessary because the drone must test routes in sequence, infer which corridors remain feasible, and possibly retreat from areas where energy consumption becomes too high. The search cost now includes not just the final path to the clinic, but all of the probing, hovering, redirection, and aborted approaches along the way.
- This example is especially good for discussing safety because exploration is no longer a harmless computational step. Each exploratory action consumes battery, may expose the drone to unsafe wind conditions, and may reduce the margin for recovery.

Why this is not just offline replanning

Even if a map exists, the task-relevant environment is still unknown because the agent does not know which routes remain viable until it interacts with them.

Classroom extension

Ask students whether the drone should prefer short unexplored paths or longer verified-safe corridors when battery is already low.

Performance considerations in online search

- Performance must be evaluated using the actual cost of behavior in the environment, not just the elegance of the final route after the fact. An online agent may eventually discover a short path to the goal yet still perform poorly if it spent substantial time and energy wandering, revisiting, or testing unproductive branches before finding it.
- This means exploration overhead is part of the metric. Every reconnaissance move, every dead-end entry, and every backtrack contributes to total path cost. In offline search, speculative expansion can be computationally expensive; in online search, it is operationally expensive.
- This perspective encourages students to ask a harder question than "Did the agent succeed?" They should ask, "How much did it pay to learn enough to succeed, and could a better exploration policy have reduced that cost?"

Keywords to stress

Exploration overhead
Backtracking
Revisits
Delayed commitment
Total traversal cost

Useful phrasing

In online search, the agent pays for uncertainty in the currency of real actions.

Worked discussion: why the discovered path may not reflect the true cost

Scenario

A campus robot eventually learns that the shortest route from Office A to Lab B is only 6 corridor segments. However, during discovery it first explores a dead-end stairwell, then a locked faculty hallway, then returns to the main lobby before finding the correct corridor.

Takeaway

The “best path” in hindsight is 6 steps, but the actual experienced cost may be 18 or more. Online performance therefore concerns the route traveled while learning, not merely the route known after learning.

Questions to ask students

- Would an agent with stronger memory necessarily do better, or could poor early choices still dominate total cost?
- If the goal is urgent, should the robot keep exploring for a potentially shorter route, or commit to the first acceptable route it has found?
- How would this discussion change if each failed corridor exploration also had a safety penalty, such as collision risk or time-critical delivery constraints?

Safety considerations in unknown environments

- Because online search requires real actions, exploration can be dangerous rather than merely expensive. An exploratory move can consume scarce battery, expose the agent to hazardous terrain, place it in a state from which retreat is difficult, or violate operational constraints such as restricted areas or fragile infrastructure.
- This means that rational online search is not only about efficiency. It is also about preserving the agent's ability to continue, recover, and remain within acceptable risk bounds. In practice, a safe route that is somewhat longer may be preferable to a shorter but poorly understood route if failure would be severe.
- An online agent should not ask only "Which move might get me closer?" It should also ask "Which move keeps me safe enough to keep searching intelligently?"

Safety themes

Reversibility
Hazard exposure
Resource depletion
Recovery margin
Operational constraints

Note

This is an applied extension of the textbook framing. The chapter foregrounds unknown-environment behavior; safety is the practical lens that makes the topic especially valuable in robotics and autonomy.

Example 3: Search-and-rescue robot in a partially collapsed structure

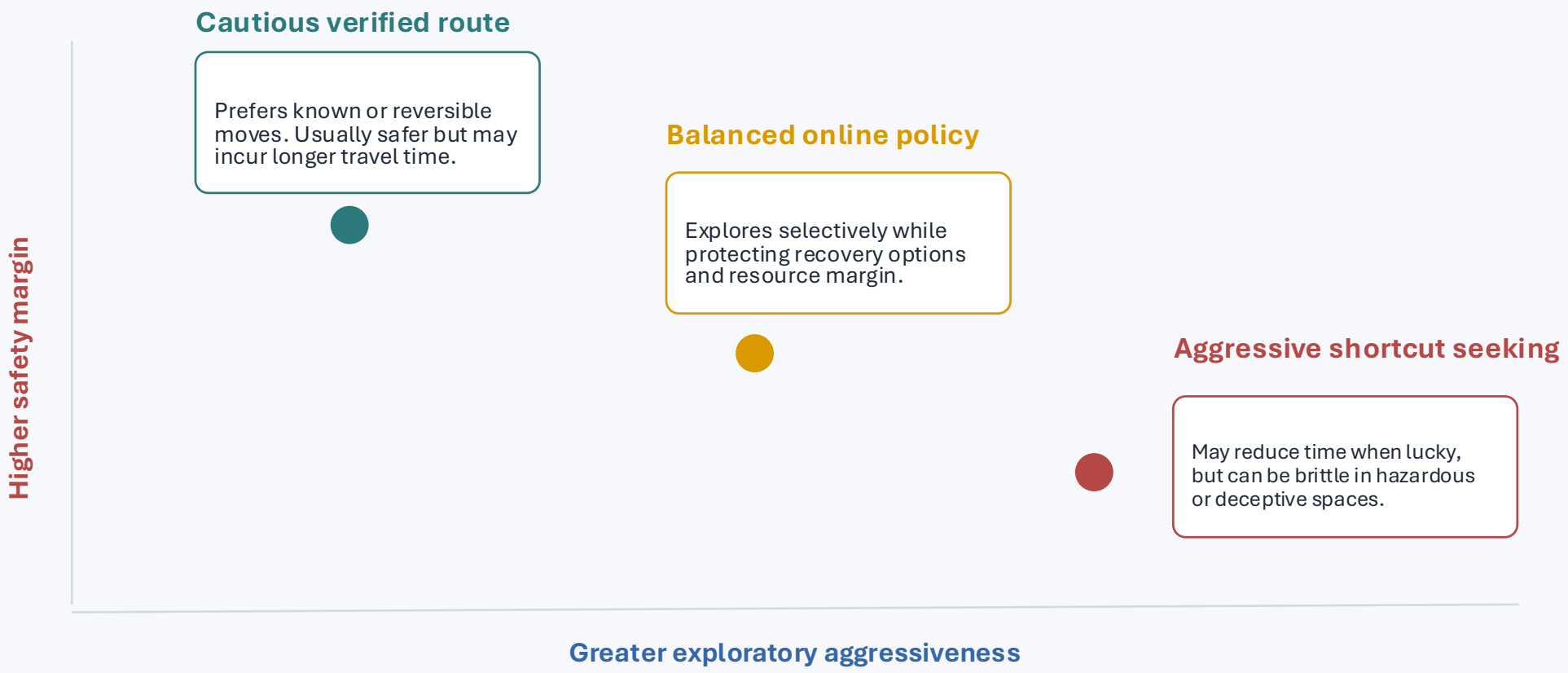
- A rescue robot is sent into a damaged building after an earthquake. It does not know which passages remain stable, which stairwells are blocked, or which rooms contain debris piles that could trap its wheels. It must therefore search online, updating its internal map while it explores.
- From a performance perspective, the robot wants to reach survivors quickly. From a safety perspective, however, aggressive exploration may be irrational because a single unstable passage could immobilize the robot and end the mission entirely. The best online behavior may therefore involve deliberately conservative probing, even if that increases travel time.
- This example helps students appreciate that online search is often about constrained rationality. The agent is not simply seeking the numerically shortest path; it is trying to achieve the mission without exhausting the very conditions required to continue acting.

Discussion angles

- Should the robot prefer verified paths over unexplored shortcuts?
- At what point does risk outweigh the value of additional exploration?
- How might mission priorities change when human life is at stake but the robot is the only available responder?

The performance–safety tradeoff

The "best" policy depends on mission stakes, reversibility of actions, confidence in local sensing, and the cost of failure.



More examples you can discuss quickly in class

Autonomous wheelchair in a new hospital wing

Knows local motion commands but not yet the layout of nurse stations, ramps, elevator availability, or temporary obstructions. Good for discussing dignity, safety, and conservative exploration around people.

Warehouse robot after layout changes

Inventory aisles have been rearranged overnight. Useful for discussing the cost of stale maps, the value of incremental remapping, and why online behavior is needed even in apparently routine environments.

Mars rover exploring rocky terrain

Excellent for discussing extreme communication delay, limited energy, hazard avoidance, and the importance of preserving the ability to recover from bad moves.

Student navigation app during a festival

Road closures and crowd barriers make the city temporarily different from the map. Good everyday example of why route knowledge may be incomplete even when the domain is familiar.

Common misconceptions to correct

- Misconception 1: “Online search just means fast search.”
  Not necessarily. The defining feature is not speed but the need to act without a complete environment model, learning relevant structure through execution.

- Misconception 2: “If the agent can observe its current state, then the problem is basically offline.”
  False. The agent may know where it currently is yet still be unable to plan globally because the rest of the environment remains undiscovered.

- Misconception 3: “Once the goal is reached, performance was good.”
  Also false. Reaching the goal does not erase the cost of excessive exploration, poor route commitment, or avoidable risk accumulated along the way.

- Misconception 4: “Safety is separate from search.”
  In practical online settings, safety is inseparable from search because exploratory moves are real interventions in the environment.

Synthesis

- Offline search is appropriate when the agent knows enough to compute a route before acting. Online search is necessary when the environment is unknown enough that the agent must move in order to learn what can be planned.
- The distinctive cost of online search is that uncertainty is paid for through real actions. Wrong turns are not merely nodes expanded in memory; they are traversed path segments, delayed progress, and sometimes additional operational risk.
- Performance therefore concerns total experienced cost, while safety concerns whether the agent preserves its ability to continue acting intelligently despite incomplete knowledge. The most rational online agent is not always the boldest one; often it is the one that explores productively without gambling away recovery options.

One-line summary for teaching

Offline search solves a known map before travel. Online search travels in order to discover enough of the map to keep solving.

Bridge forward

This topic prepares students for richer discussions of exploration, learning, replanning, and autonomous behavior in dynamic settings.

Things to Ponder

“If failure has a very high cost, should the agent still prefer information-gathering actions, or should it settle quickly for a merely acceptable route?”

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